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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

models = {
    "Logistic Regression": LogisticRegression(max_iter=200),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Naive Bayes": GaussianNB()
}

print("CLASSIFICATION ALGORITHM COMPARISON")
print("-----")

for name, model in models.items():

    if name in ["Logistic Regression", "KNN"]:
        model.fit(X_train_scaled, y_train)
        prediction = model.predict(X_test_scaled)

    else:
        model.fit(X_train, y_train)
        prediction = model.predict(X_test)

    accuracy = accuracy_score(y_test, prediction)

    print("\nAlgorithm:", name)

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print("Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

print("\nComparison Completed Successfully")

    CLASSIFICATION ALGORITHM COMPARISON
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...

    Algorithm: Logistic Regression
    Accuracy: 0.9333333333333333
    Accuracy Percentage: 93.33333333333333 %

    Algorithm: KNN
    Accuracy: 0.9333333333333333
    Accuracy Percentage: 93.33333333333333 %

    Algorithm: Decision Tree
    Accuracy: 0.9333333333333333
    Accuracy Percentage: 93.33333333333333 %

    Algorithm: Naive Bayes
    Accuracy: 0.9666666666666667
    Accuracy Percentage: 96.66666666666667 %
```

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